Po-Sheng Cheng

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Portfolio: https://bencer3283.github.io/art/

Master of Industrial Design, Rhode Island School of Design, exp. Jun. 2026
Cross-enrolled at School of Engineering, Brown University
B.Sc. Bio-Mechatronics Engineering, National Taiwan University, Jan 2023

Experiences

Faculty of Record, Dept. of Industrial Design, RISD, Jan. - Feb. 2026 — Providence, RI

- Proposed, awarded and taught an introductory studio class (ID1531) in physical computing and embedded system development. Employed theoretical lectures, demo, and hands-on workshop to cover topic like microcontroller, circuit theory, power rails, binary arithmetic, serial communication, HTTP, and machine learning.
- Blended technical skill with discussion on the societal impact of technology with framework like Science and Technology Studies.

Design Technologist Intern, Smart Design, Jun. - Aug. 2025 — Brooklyn, NY

- Developed embedded firmware (STM32) for prototyping the HMI of a domestic product, while working with UX design to test and fine-tune user interaction flow and hardware user interface.
- Developed testing units for motor and pneumatic mechanisms in the product, including the electrical hardware, embedded firmware and host software. (PyQt)

Product Development Apprentice, Google, Jun. - Sep. 2024 — Taipei, Taiwan

- Led a team of 5 to design a Hardware-as-a-Service platform for managing lost items in public facilities. Laid out the system architecture of the hardware kiosk, user-facing app and web services.
- My prototypes of Raspberry Pi and ESP32 hardware, a React-based website, a mobile app and a RESTful API service drove the development direction from a single hardware product to an integrated platform.

Electro-mechanical Engineering Intern, Logitech, Feb. - Jun. 2022 — Hsinchu, Taiwan

- Created a brand new breed of keyboard technology by conducting quantitative user research with 50 interviewees and innovating an Arduino microcontroller-based prototype to control actuators like stepper motor and electromagnet. Also lots of CAD (Creo) and SLA/FDM prototyping on the mechanical design.

- The prototype received widespread praise across many departments (pm, engineering, design) in the company while I worked with them to support guiding the direction of product development.

Publications

TeleSHift: Telexisting TUI for Physical Collaboration & Interaction

- Architected an ESP32-based embedded system to implement a shape-transforming device with a 3D tangible user interface.
- Designed the robust interactive mechanisms (SolidWorks). My design for manufacturability (DFM) improvements also slashed assembling time by 80% during demo.

Awards

RISD Grad Common Grant 2025

awarded by the Rhode Island School of Design for the continuation of the MITM Router project.

Best Demo Paper at 2022 ACM Ubicomp conference

awarded for our paper TeleSHift: Telexisting TUI for Physical Collaboration & Interaction.

College Student Research Creativity Award

awarded by the National Science and Technology Council of Taiwan for a research project building novel spectral mapping technologies.